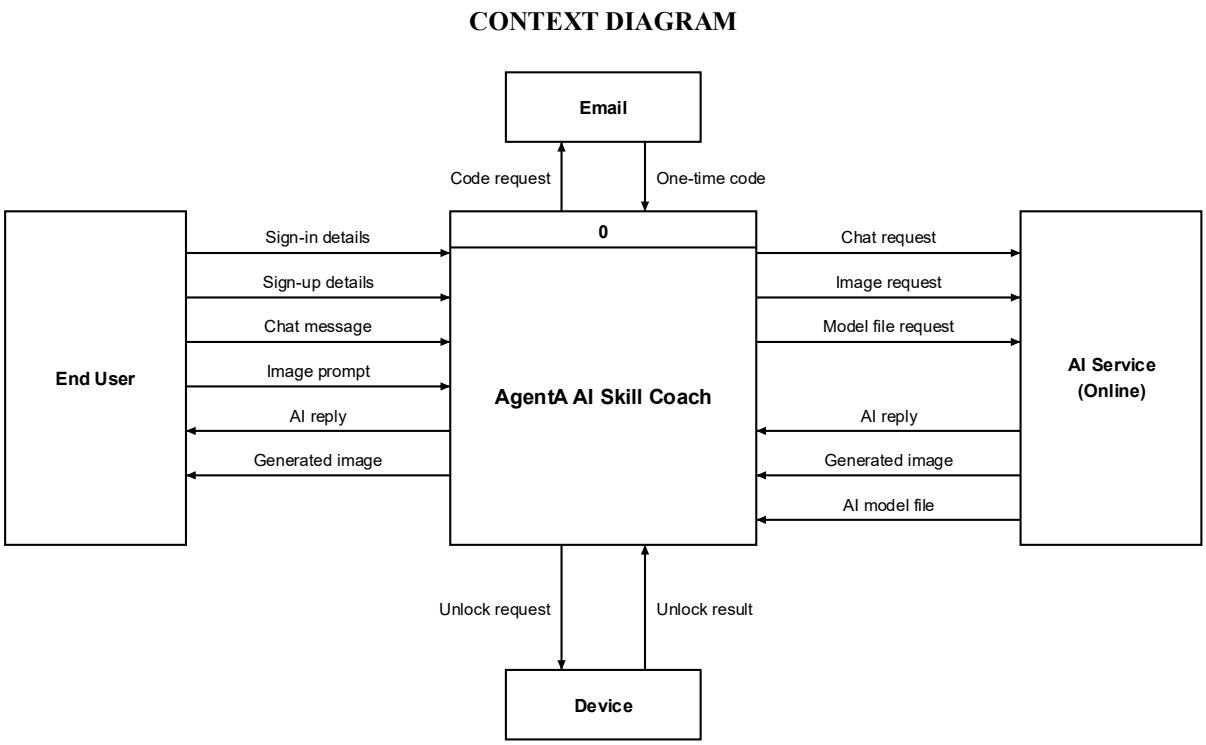


Figure 1. Context Diagram of the AgentA System

Narrative Flow of Data

Figure 1 shows the AgentA AI Skill Coach system as one process, numbered 0, together with the four outside parties it exchanges data with. The End User signs in or creates an account by sending sign-in details and sign-up details to process 0, and while using the app the same user sends chat messages and image prompts. Process 0 answers the End User with AI replies and generated images. When the phone is online, process 0 sends chat requests, image requests, and model file requests to the AI Service (Online), and the AI Service returns AI replies, generated images, and the AI model file in response. When a password is forgotten, process 0 sends a code request to Email, and Email returns a one-time code that the user types back into the app to set a new password. Process 0 also asks the Device to check the user's fingerprint or face by sending an unlock request, and the Device answers with the unlock result that lets the user back into the app. Every action shown here begins with the End User, and every answer reaches the End User through process 0.

LEVEL 1 DFD (DIAGRAM 0)

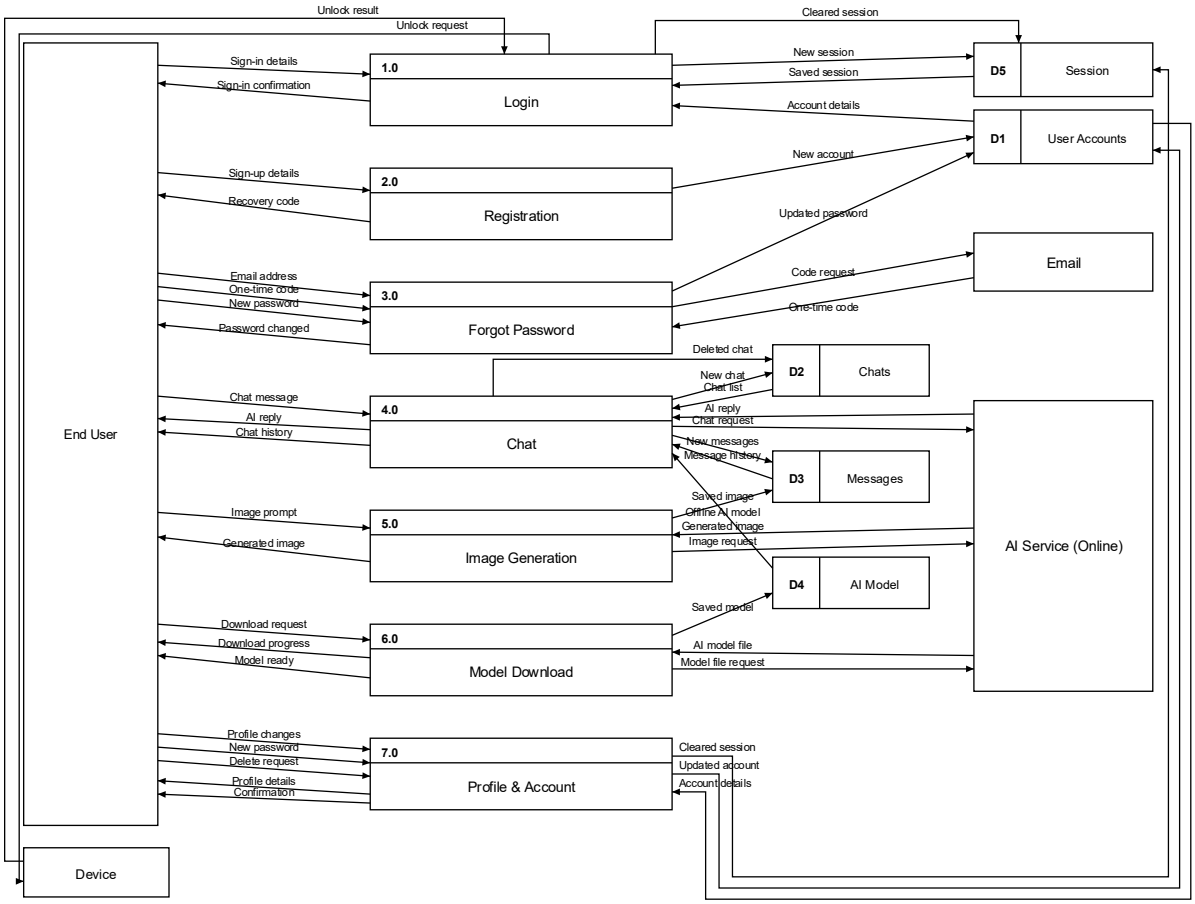


Figure 2. Level 1 DFD (Diagram 0) of the AgentA System

Narrative Flow of Data

To use AgentA, the End User signs in through process 1.0 Login, which checks account details in D1 User Accounts and asks the Device for a biometric unlock; the active session is kept in D5 Session, can be restored while it is still saved, and is cleared again when the user signs out. New users go through 2.0 Registration, which stores a new account in D1 User Accounts and returns a one-time recovery code to the End User. If a password is forgotten, 3.0 Forgot Password sends a code request to Email, the End User enters the one-time code and a new password, and the updated password is written back to D1 User Accounts. Signed-in users chat through 4.0 Chat, which saves every new chat in D2 Chats and every new message in D3 Messages; deleting a chat removes it from D2 Chats; when the connection is online the chat request goes to the AI Service (Online), which returns the AI reply, and when offline the reply comes from the offline AI model held in D4 AI Model. Image requests pass through 5.0 Image Generation, which asks the AI Service for the picture and saves the finished image in D3 Messages so it stays part of the chat. On first use, 6.0 Model Download fetches the AI model file from the AI Service, shows download progress to the End User, and saves the model in D4 AI Model until it is ready. Finally, 7.0 Profile & Account sends profile changes and a new password to D1 User Accounts, and confirms every change back to the End User; deleting the account also clears the saved session from D5 Session.

LEVEL 2 DFD PROCESS 1.0 (LOGIN)

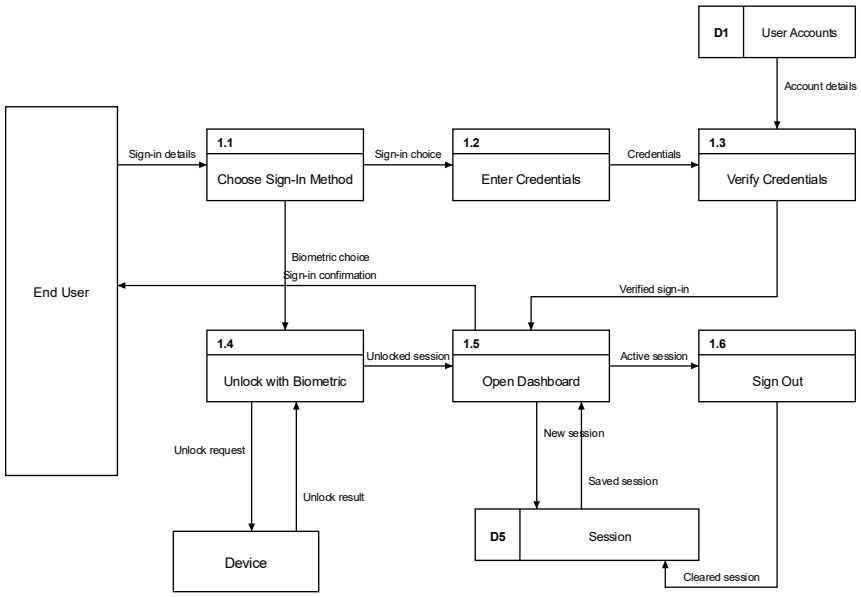


Figure 3. Level 2 DFD Process 1.0 (Login)

Narrative Flow of Data

When the End User opens the app, process 1.1 Choose Sign-In Method starts the login by offering either a password sign-in or a biometric unlock. Choosing the password route moves to 1.2 Enter Credentials, where the user types the sign-in details, and the typed credentials flow into 1.3 Verify Credentials. Verification checks the account details kept in D1 User Accounts, and once the details match, a verified sign-in moves on to 1.5 Open Dashboard. Choosing the biometric route goes to 1.4 Unlock with Biometric, which sends an unlock request to the Device and continues only after the Device returns a successful unlock result. When the dashboard opens, 1.5 Open Dashboard shows the sign-in confirmation back to the End User and writes a new session into D5 Session, while an earlier saved session can also be read from D5 Session when the app starts again. Finally, 1.6 Sign Out ends the active session and clears the saved session from D5 Session.

LEVEL 2 DFD PROCESS 2.0 (REGISTRATION)

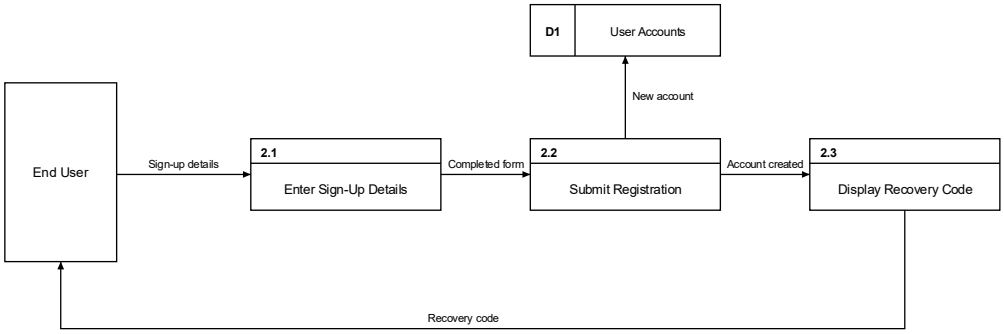


Figure 4. Level 2 DFD Process 2.0 (Registration)

Narrative Flow of Data

Registration starts when the End User opens the sign-up page; process 2.1 Enter Sign-Up Details takes the sign-up details that the user types, such as an email address and a password. The completed form flows into 2.2 Submit Registration, which creates the account and sends the new account into D1 User Accounts. Once the account exists, the account created message moves to 2.3 Display Recovery Code, which shows the one-time recovery code on the screen. The recovery code flows back to the End User, who is asked to keep it safe for recovering the account later. With the account safely stored in D1 User Accounts and the recovery code in the user's hands, the registration is complete.

LEVEL 2 DFD PROCESS 3.0 (FORGOT PASSWORD)

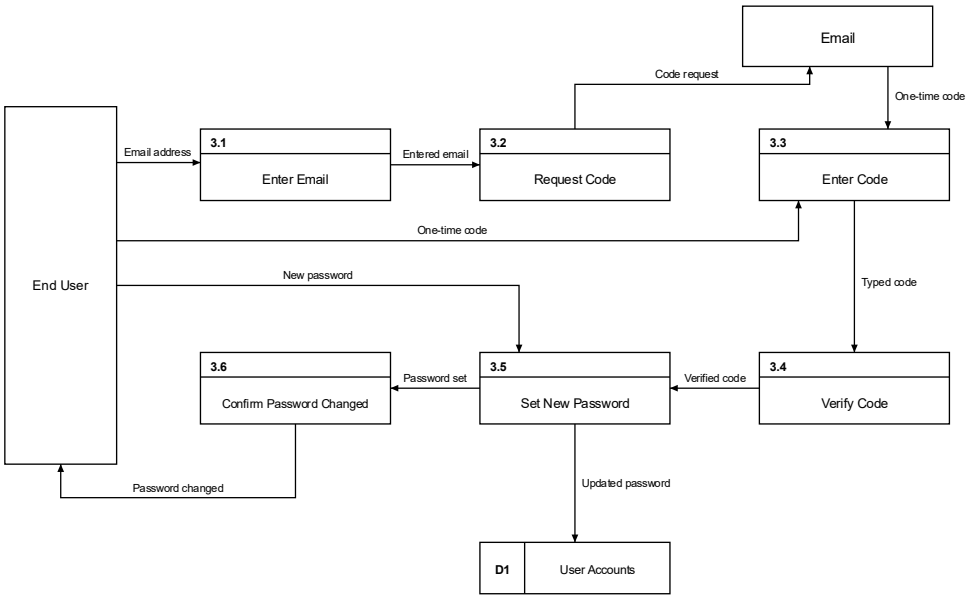


Figure 5. Level 2 DFD Process 3.0 (Forgot Password)

Narrative Flow of Data

When the End User forgets a password, process 3.1 Enter Email takes the email address typed on the reset screen. The entered email flows into 3.2 Request Code, which sends a code request to Email and asks for a one-time code to be delivered. The Email entity sends the one-time code to 3.3 Enter Code, and the End User types that same one-time code to continue. The typed code flows into 3.4 Verify Code, which checks the code and passes a verified code on to 3.5 Set New Password. The End User types a new password into 3.5, which sends the updated password into D1 User Accounts. Finally, the password set result moves to 3.6 Confirm Password Changed, which shows the password changed confirmation back to the End User.

LEVEL 2 DFD PROCESS 4.0 (CHAT)

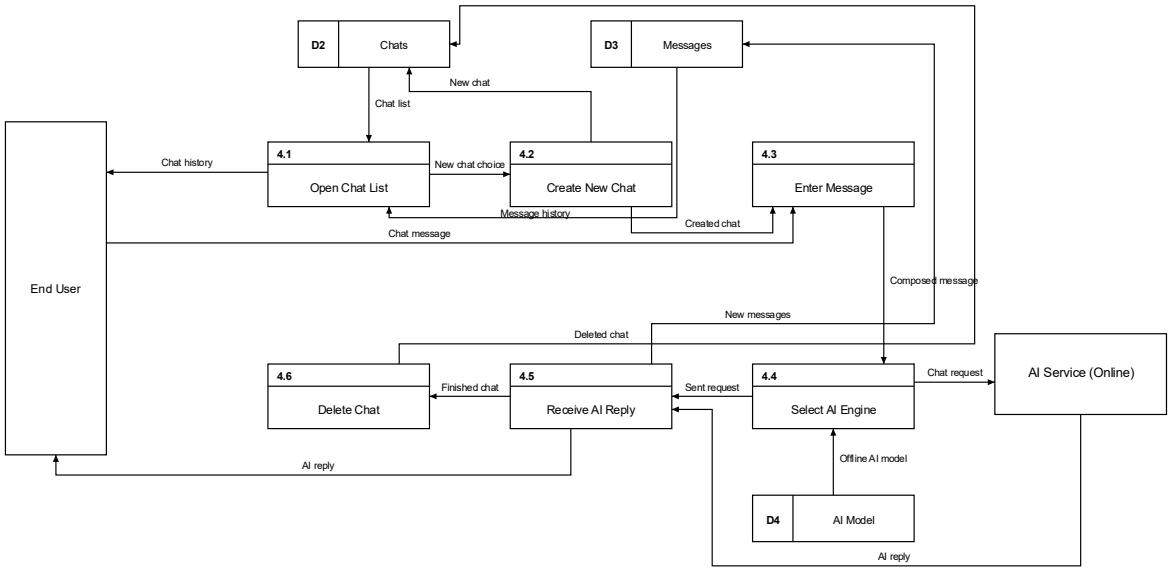


Figure 6. Level 2 DFD Process 4.0 (Chat)

Narrative Flow of Data

When the End User opens the chat area, process 4.1 Open Chat List reads the chat list from D2 Chats and the message history from D3 Messages, and shows the chat history back to the End User. Creating a new chat through 4.2 Create New Chat saves a new chat in D2 Chats, and the created chat moves on to 4.3 Enter Message. There the End User types a chat message, and the composed message flows into 4.4 Select AI Engine. Process 4.4 chooses the AI engine, using the offline AI model stored in D4 AI Model when the device is offline, and sends the chat request to the AI Service (Online) when the device is online. The AI Service returns the AI reply to 4.5 Receive AI Reply, which shows the AI reply to the End User and saves the conversation as new messages in D3 Messages. Finally, 4.6 Delete Chat removes a chat from D2 Chats when the user deletes it.

LEVEL 2 DFD PROCESS 5.0 (IMAGE GENERATION)

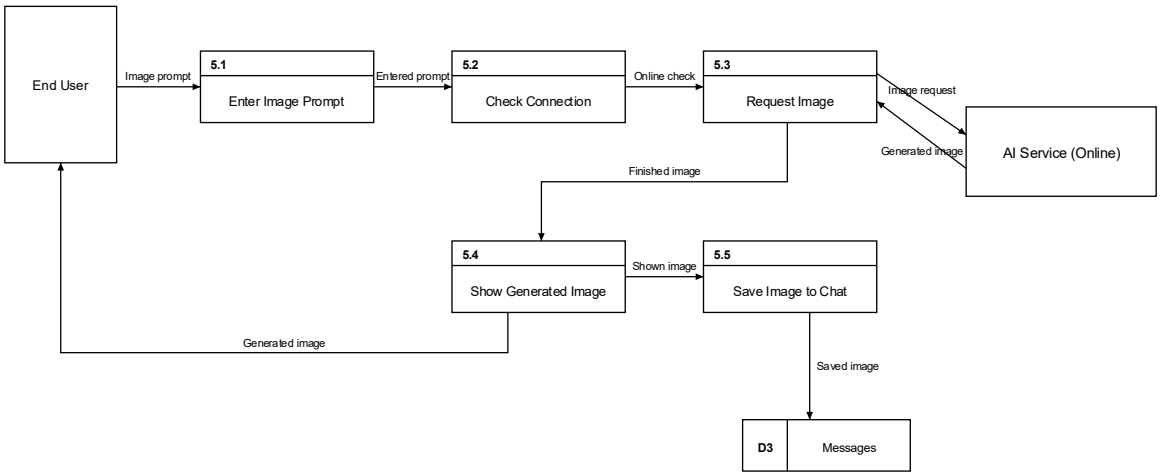


Figure 7. Level 2 DFD Process 5.0 (Image Generation)

Narrative Flow of Data

Image generation starts when the End User types an image prompt, and process 5.1 Enter Image Prompt takes the typed prompt. The entered prompt moves to 5.2 Check Connection, which checks whether the device is online and passes an online check to 5.3 Request Image. Process 5.3 sends the image request to the AI Service (Online), which generates the picture and returns the generated image to 5.3. The finished image flows into 5.4 Show Generated Image, which shows the picture to the End User. When the user keeps it, the shown image moves to 5.5 Save Image to Chat, which saves it as a saved image in D3 Messages so it stays part of the chat.

LEVEL 2 DFD PROCESS 6.0 (MODEL DOWNLOAD)

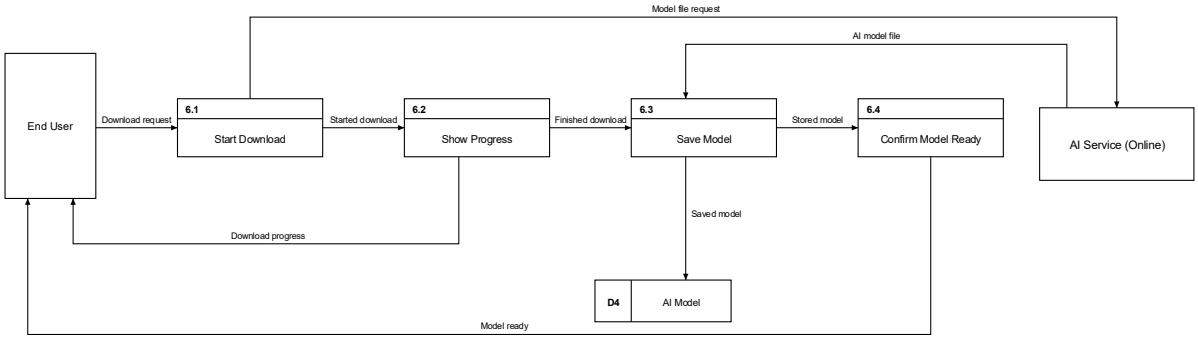


Figure 8. Level 2 DFD Process 6.0 (Model Download)

Narrative Flow of Data

The download begins when the End User taps to start it, and process 6.1 Start Download takes the download request. Process 6.1 sends the model file request to the AI Service (Online), which returns the AI model file to 6.3 Save Model. While the file arrives, the started download moves to 6.2 Show Progress, which shows download progress back to the End User. The finished download passes from 6.2 to 6.3 Save Model, which saves the model as a saved model in D4 AI Model. Finally, the stored model moves to 6.4 Confirm Model Ready, which tells the End User that the model is ready, and the app can then keep working offline.

LEVEL 2 DFD PROCESS 7.0 (PROFILE & ACCOUNT)

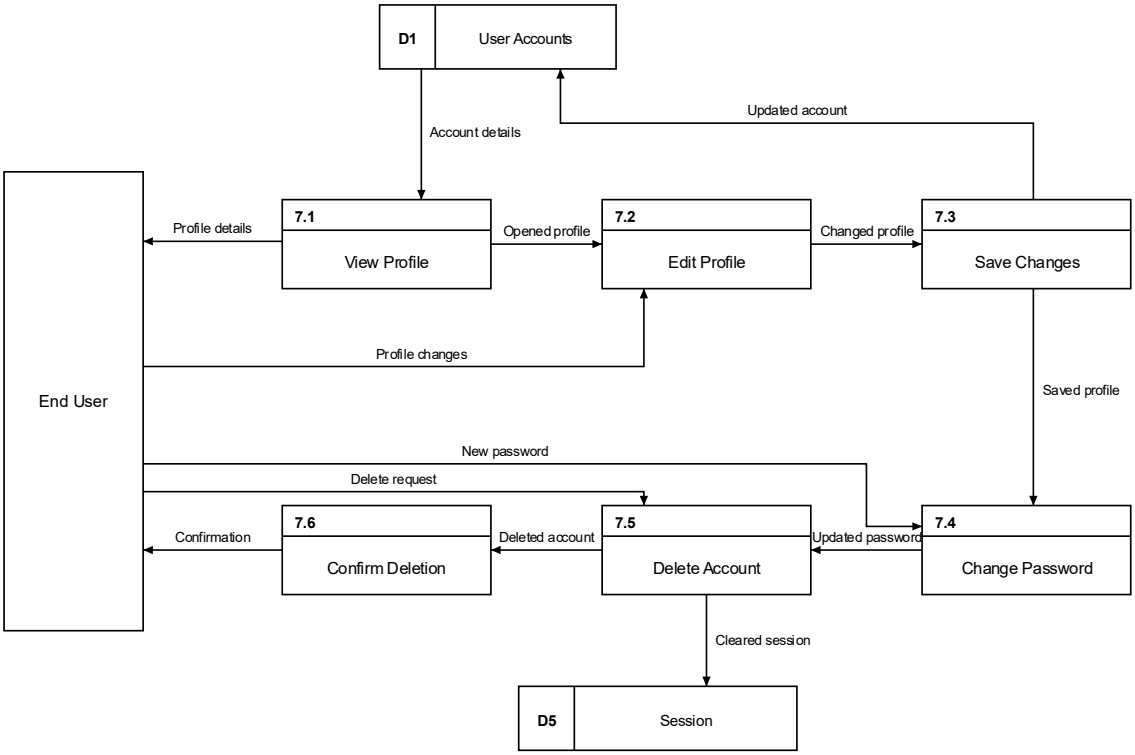


Figure 9. Level 2 DFD Process 7.0 (Profile & Account)

Narrative Flow of Data

Profile management starts when process 7.1 View Profile reads the account details kept in D1 User Accounts and shows the profile details to the End User. The opened profile moves to 7.2 Edit Profile, where the End User changes the profile details and sends the profile changes. The changed profile flows into 7.3 Save Changes, which writes the updated account back into D1 User Accounts. The saved profile continues to 7.4 Change Password, where the End User enters a new password to update. The updated password moves to 7.5 Delete Account, where the End User can send a delete request; when the account is deleted, the cleared session is removed from D5 Session. Finally, the deleted account message goes to 7.6 Confirm Deletion, which shows the confirmation to the End User.